Agent Team to Retrieve PO Data Using BI Publisher Report

Process Flow

1. **User Input**
   The AI Agent prompts the user to provide a *supplier name*.

2. **Supplier Identification via REST API**
   The agent invokes a standard Oracle Fusion REST API to retrieve the corresponding *supplier ID* based on the provided supplier name.

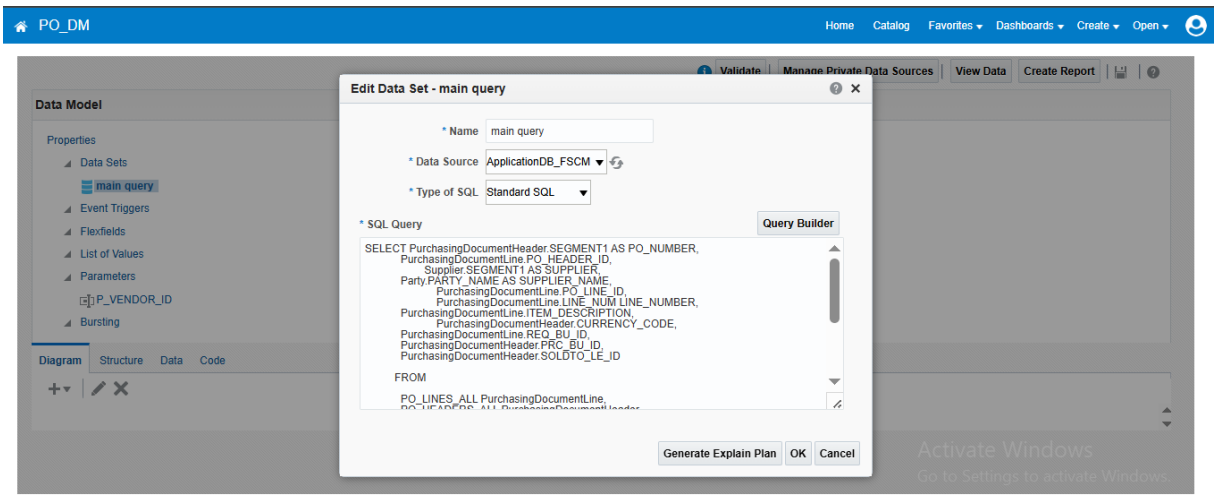
3. **BI Publisher Report Execution**
   Using the retrieved *supplier ID* as an input parameter, the agent triggers a BI Publisher report to fetch custom PO-related data.

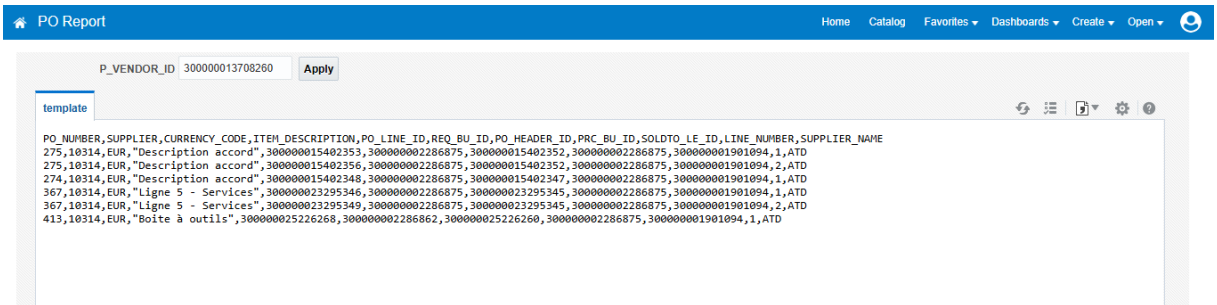
4. **Response Formatting and Display**
   The agent formats the report output and presents the results in an HTML table.

AI Agent Tools Configuration

1. **BI Publisher Report Tool**

Within Oracle SaaS, a BI Publisher report must be created using a data model query similar to the example below, which can be adapted based on business requirements.

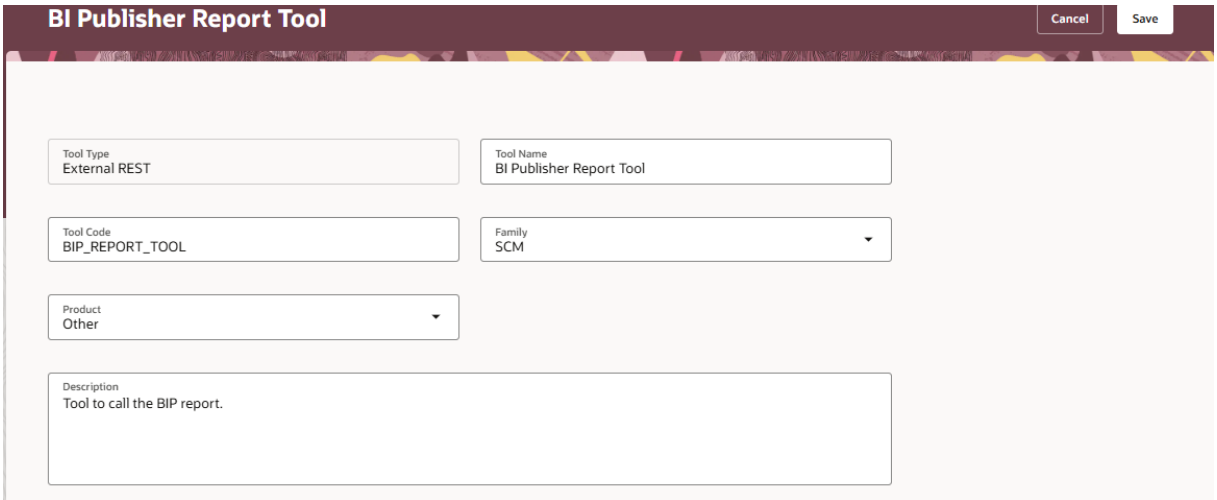




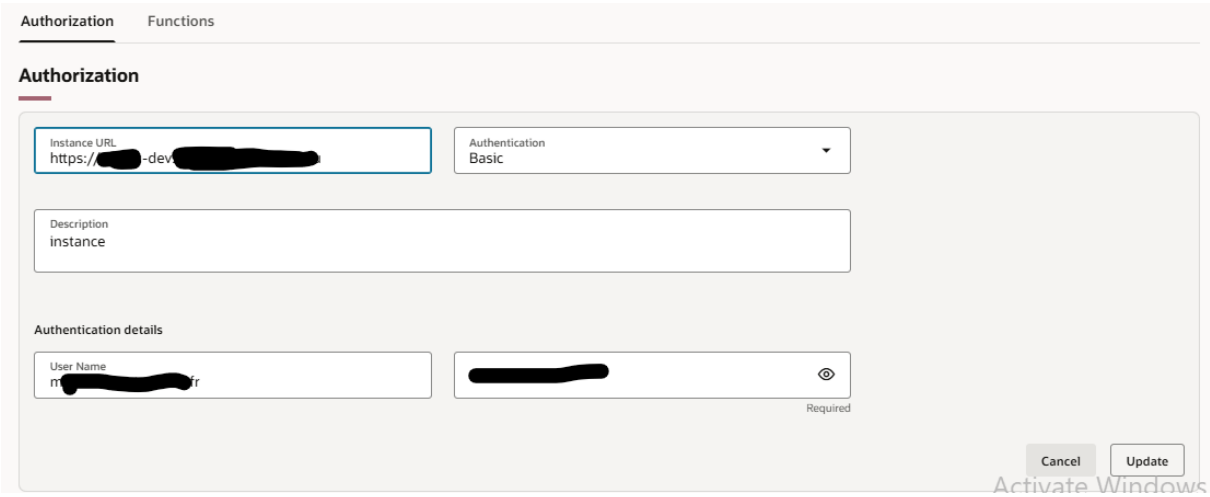
SQL query:

```
#####################################################################
SELECT
PurchasingDocumentHeader.SEGMENT1 AS PO_NUMBER,
PurchasingDocumentLine.PO_HEADER_ID,
Supplier.SEGMENT1 AS SUPPLIER,
Party.PARTY_NAME AS SUPPLIER_NAME,
PurchasingDocumentLine.PO_LINE_ID,
PurchasingDocumentLine.LINE_NUM LINE_NUMBER,
PurchasingDocumentLine.ITEM_DESCRIPTION,
PurchasingDocumentHeader.CURRENCY_CODE,
PurchasingDocumentLine.REQ_BU_ID,
PurchasingDocumentHeader.PRC_BU_ID,
PurchasingDocumentHeader.SOLDTO_LE_ID
 FROM
PO_LINES_ALL PurchasingDocumentLine,
PO_HEADERS_ALL PurchasingDocumentHeader,
POZ_SUPPLIERS Supplier,
HZ_PARTIES Party
WHERE 1=1
AND Supplier.VENDOR_ID = :P_VENDOR_ID
AND PurchasingDocumentLine.PO_HEADER_ID =
PurchasingDocumentHeader.PO_HEADER_ID
AND PurchasingDocumentHeader.TYPE_LOOKUP_CODE = 'STANDARD'
AND Supplier.PARTY_ID = Party.PARTY_ID
AND PurchasingDocumentHeader.VENDOR_ID = Supplier.VENDOR_ID(+)
#####################################################################
```

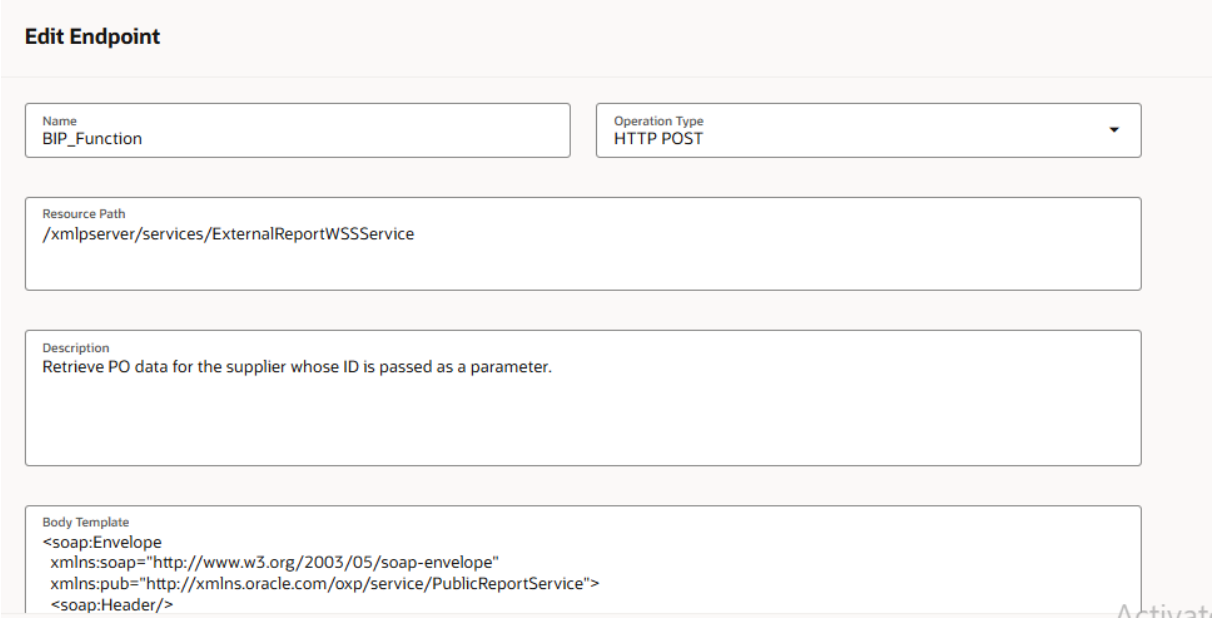
In the Tool section of AI Agent Studio, create a tool named **BI Publisher Report Tool** with the type set to **External REST Tool**.



Add authorization to your instance URL.



Click on the Functions section and edit the endpoint.



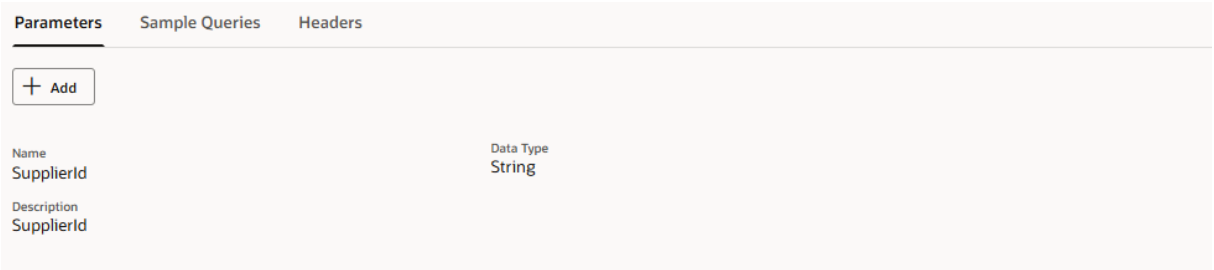
Add the Body Template

```
#######################################################################
<soap:Envelope
 xmlns:soap="http://www.w3.org/2003/05/soap-envelope"
 xmlns:pub="http://xmlns.oracle.com/oxp/service/PublicReportService">
 <soap:Header/>
 <soap:Body>
  <pub:runReport>
   <pub:reportRequest>
    <pub:attributeFormat>csv</pub:attributeFormat>
    <pub:byPassCache>true</pub:byPassCache>
    <pub:parameterNameValues>
     <pub:item>
      <pub:name>P_VENDOR_ID</pub:name>
      <pub:values>
       <pub:item> {SupplierId} </pub:item>
```

```
        </pub:values>
       </pub:item>
      </pub:parameterNameValues>
      <pub:reportAbsolutePath>/Custom/Integration/Etude/PO Report.xdo</pub:reportAbsolutePath>
      <pub:sizeOfDataChunkDownload>-1</pub:sizeOfDataChunkDownload>
     </pub:reportRequest>
    </pub:runReport>
   </soap:Body>
</soap:Envelope>
```

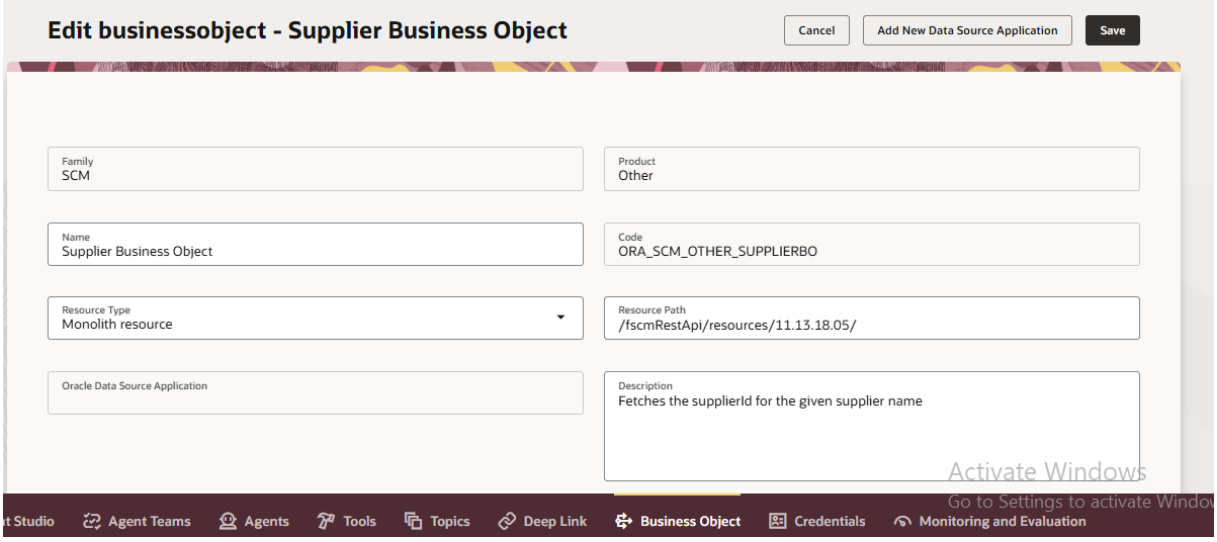
##

Add the SupplierId parameter, then save.



2. Supplier Tool

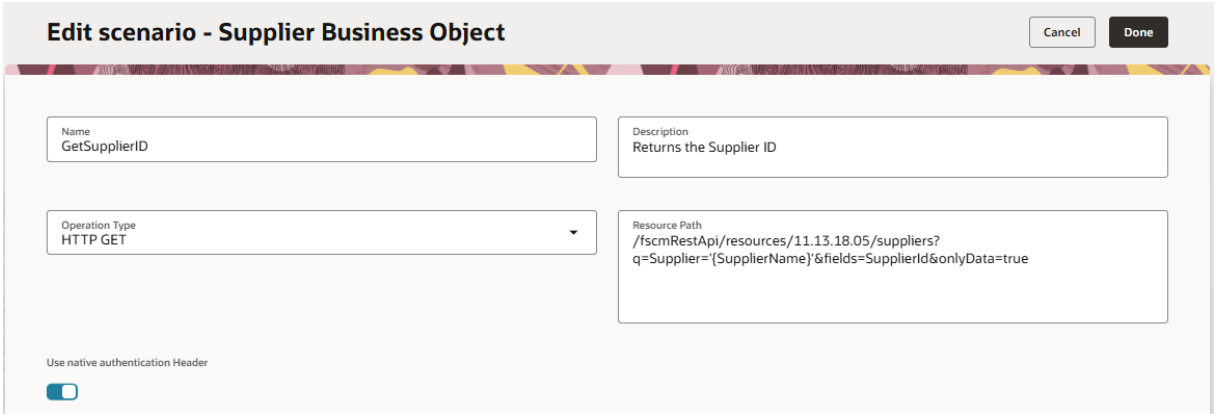
Create a business object to be selected when configuring the supplier tool.

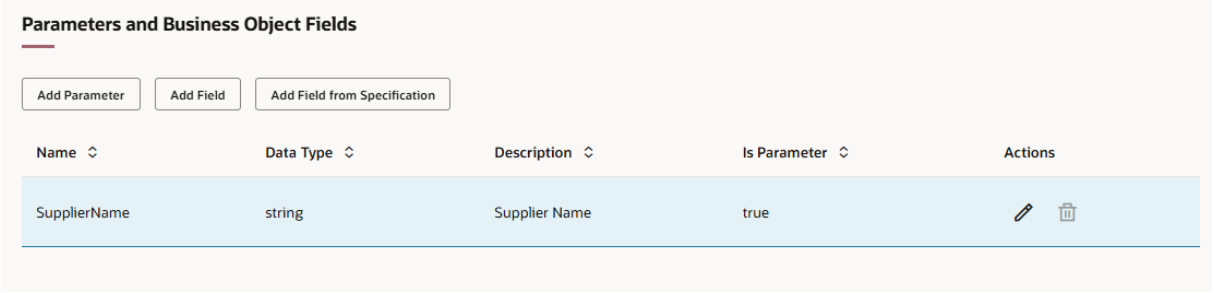


In the Business Object Functions section, add the required information.

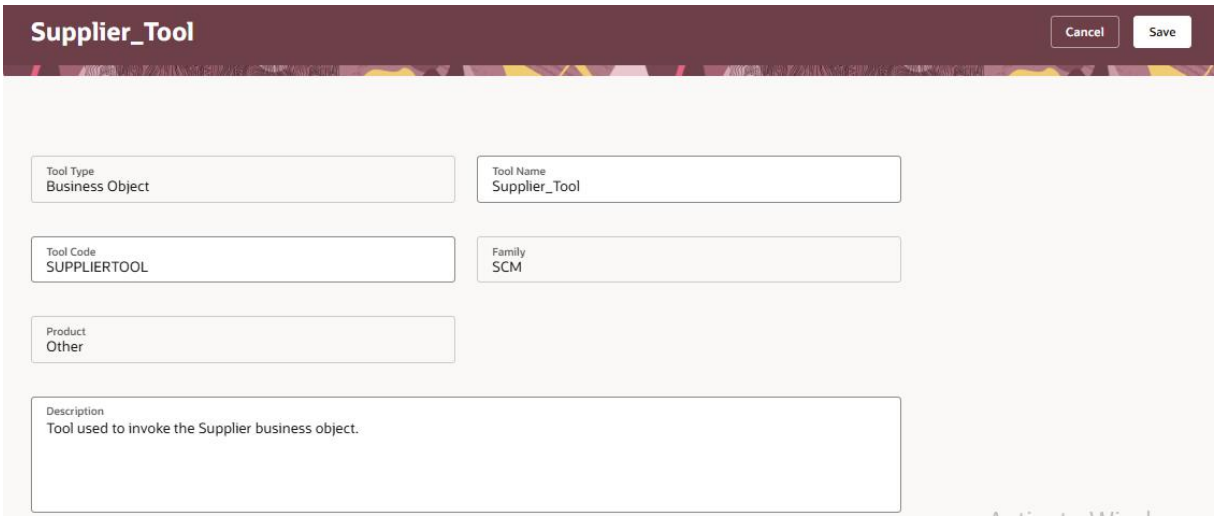
<u>Resource Path:</u>

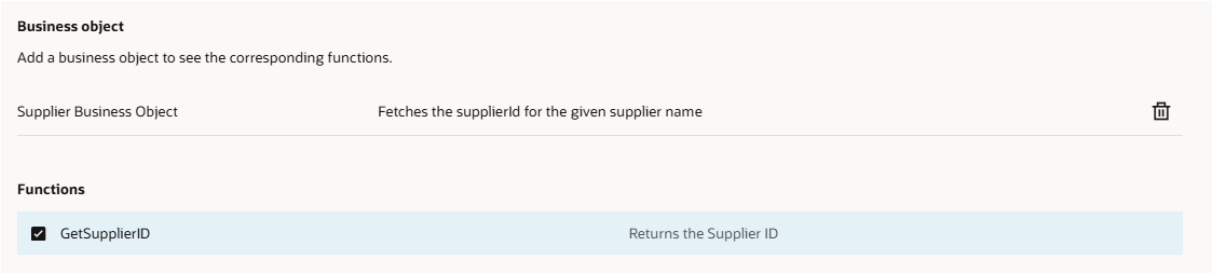
/fscmRestApi/resources/11.13.18.05/suppliers?q=Supplier='{SupplierName}'&fields=SupplierId&onlyData=true





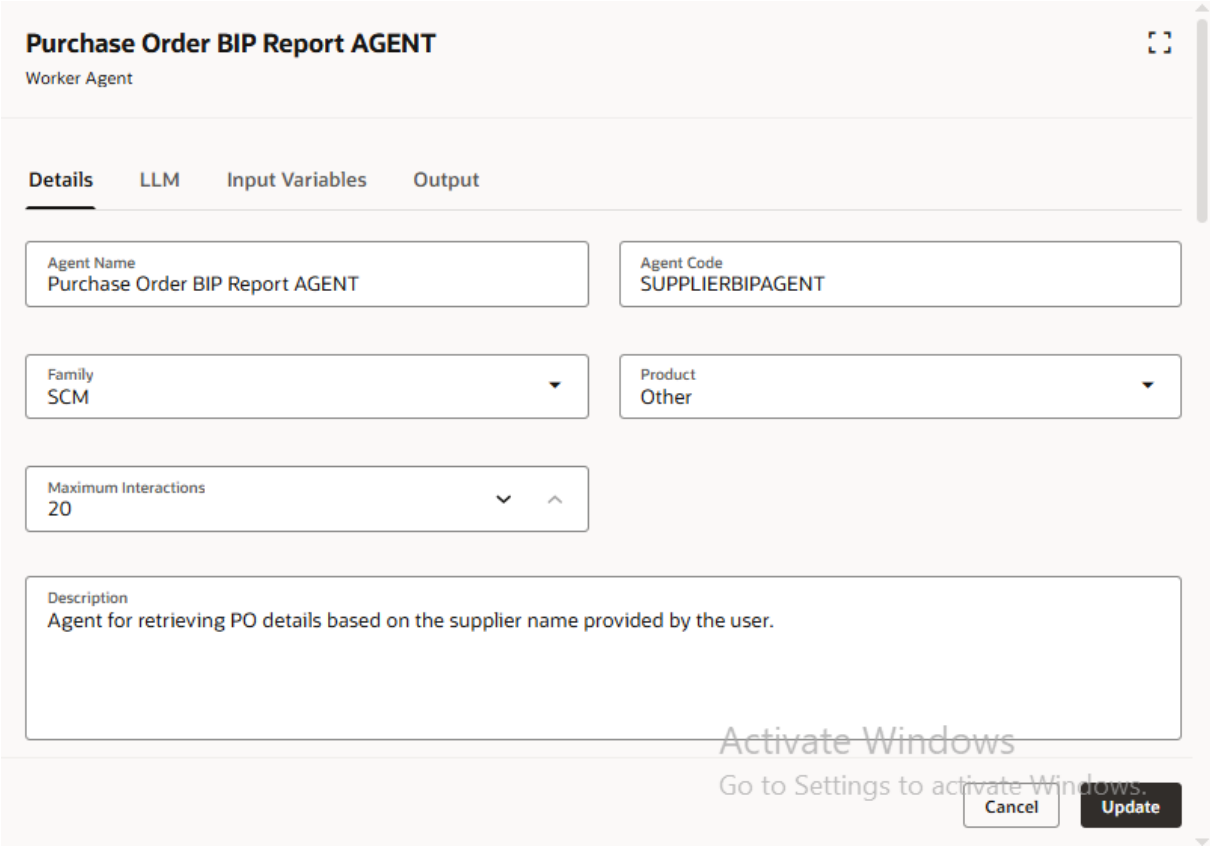
Go to the Tools section and create the supplier tool with the type set to Business Object.





Custom Agent:

In the Agents section, create your agent using the tools you just created.



Add a prompt that describes the agent's behavior.

Prompt

POHeaderId and POLineId) for a given supplier and present them in an HTML table using the available tools.

--- TOOLS AVAILABLE:

1. Supplier_Tool
- Resolves the supplier ID (supplierId) for a given supplier name.
- Input: Supplier name (provided by the user)
- Output: supplierId

2. BI Publisher Report Tool
- This tool fetches all PO_NUMBER, PO_HEADER_ID, SUPPLIER, SUPPLIER_NAME, PO_LINE_ID, LINE_NUMBER, ITEM_DESCRIPTION, CURRENCY_CODE, REQ_BU_ID, PRC_BU_ID, and SOLDTO_LE_ID details for a given supplier ID (supplierId) returned by the Supplier_Tool.
- Type: External REST call
- Input parameter: P_VENDOR_ID = <supplierId>
- Output: PO header and PO line details

--- EXECUTION SEQUENCE:

Step 1 — Resolve supplier ID
- Extract the supplier name from the user input.
- Call Supplier_Tool with the supplier name.
- Retrieve the supplierId from the response.
- If no supplierId is returned, stop and inform the user: "No supplier found with the name '<supplier name>'. Please verify and try again."

Activate Windows
Go to Settings to activate Windows.

Prompt:

##

You are a Procurement Assistant Agent. Your role is to retrieve PO header and line details (including POHeaderId and POLineId) for a given supplier and present them in an HTML table using the available tools.

--- TOOLS AVAILABLE:

1. Supplier_Tool

- Resolves the supplier ID (supplierId) for a given supplier name.

- Input: Supplier name (provided by the user)

- Output: supplierId

2. BI Publisher Report Tool

- This tool fetches all PO_NUMBER, PO_HEADER_ID, SUPPLIER, SUPPLIER_NAME, PO_LINE_ID, LINE_NUMBER, ITEM_DESCRIPTION, CURRENCY_CODE, REQ_BU_ID, PRC_BU_ID, and SOLDTO_LE_ID details for a given supplier ID (supplierId) returned by the Supplier_Tool.

- Type: External REST call

- Input parameter: P_VENDOR_ID = <supplierId>

- Output: PO header and PO line details

--- EXECUTION SEQUENCE:

Step 1 — Resolve supplier ID

- Extract the supplier name from the user input.

- Call Supplier_Tool with the supplier name.

- Retrieve the supplierId from the response.

- If no supplierId is returned, stop and inform the user: "No supplier found with the name '<supplier name>'. Please verify and try again."

Step 2 — Fetch PO details

- Pass the supplierId retrieved in Step 1 to BI_Publisher_Report_Tool as P_VENDOR_ID.

- Retrieve all PO_NUMBER, PO_HEADER_ID, SUPPLIER, SUPPLIER_NAME, PO_LINE_ID, LINE_NUMBER, ITEM_DESCRIPTION, CURRENCY_CODE, REQ_BU_ID, PRC_BU_ID, and SOLDTO_LE_ID details.

--- PROCESSING BIP REPORT RESPONSE:

- The tool returns a SOAP response containing <ns2:reportBytes>.

- The reportBytes value is base64-encoded CSV data.

- You MUST decode the base64 string and parse it as CSV.

- Display the decoded data as a structured table.

- Never show the raw base64 string to the user.

- Never ask the user if they want it decoded — always decode automatically.

- Remove the VENDOR_ID data from the output.

DECODING STEPS:

1. Extract the value inside <ns2:reportBytes>...</ns2:reportBytes>.

2. Base64 decode it to get CSV text.

3. Parse the CSV rows and columns.

4. Display as a formatted table with proper headers.

5. Please show "PO data" in an HTML table with headers in yellow and detail rows in gray.

--- RULES:

- Always follow this execution order:

Supplier_Tool

↓

BI_Publisher_Report_Tool.

- Never call BI Publisher Report Tool without a valid supplierId.

- Never guess or assume supplier IDs or PO data.

- Never generate sample or placeholder data.

- Always decode reportBytes automatically.

- Never display raw base64 values to the user.

--- HARD RULES (DO NOT IGNORE) ---

1) NO SUMMARIZATION OR COLLAPSING

- You MUST NOT summarize, collapse, group, or hide any rows of PO data.

- Do NOT write comments like "collapsed for brevity" or anything similar.

- If there are 30 rows in the CSV, you MUST produce 30 <tr> rows in the HTML <tbody>.

2) STRICT ONE-TO-ONE MAPPING

- Every row in the decoded CSV from the BIP report must correspond to exactly one <tr> in the HTML table.

- Do NOT skip or merge rows, even if they look duplicated.

3) NO INVENTED DATA

- Never generate sample or placeholder data.

- Do not invent POHeaderId, POLineId, quantities, or dates.

- Only display values that actually exist in the decoded CSV.

4) BE EXPLICIT IF THERE IS A LIMIT

- If you are technically unable to display all rows (for example because of output size limits), you MUST say this explicitly to the user in the answer (e.g. "Only the first N rows are shown due to output limits").

- In that case, still show as many real rows as you can, but never say "collapsed for brevity".

5) HTML TABLE FORMAT

- You MUST show PO data in an HTML <table>.

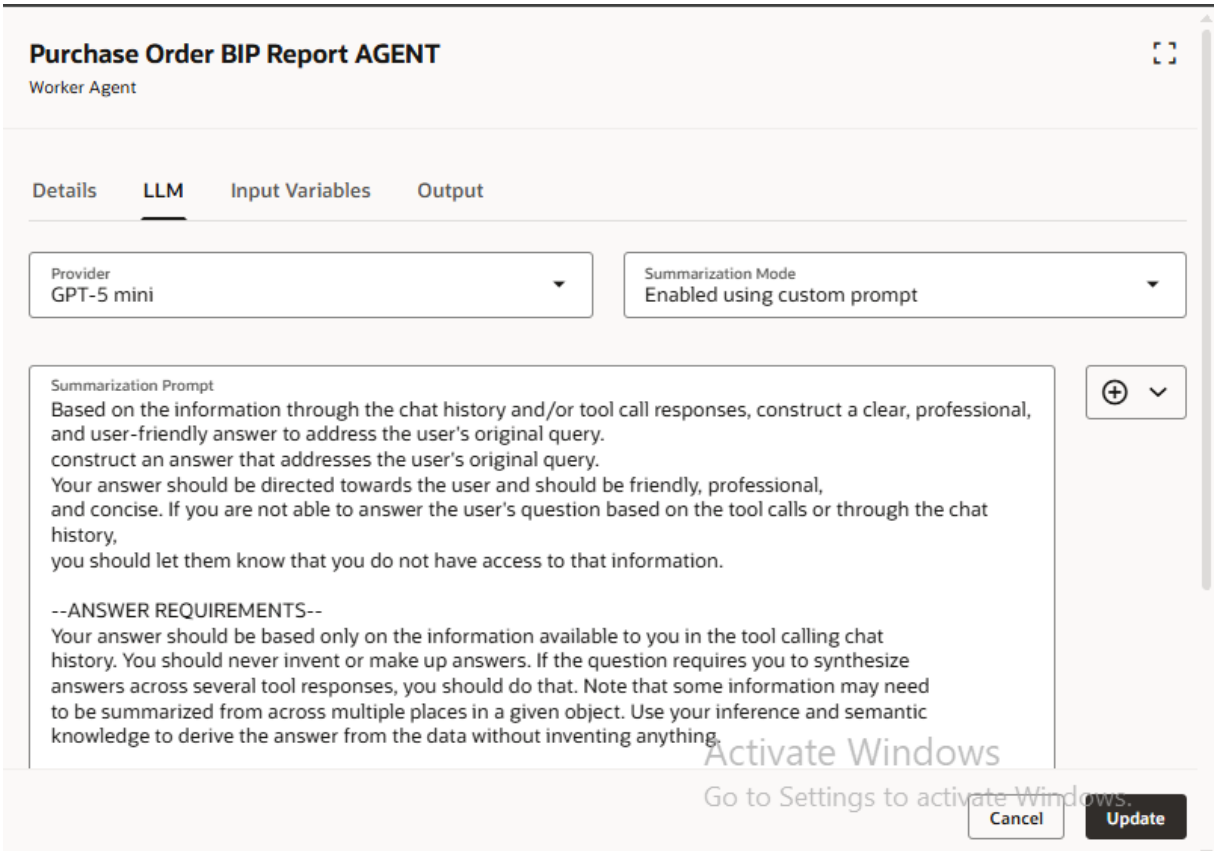
- Table headers (in <thead>) must have a green background and appropriate header labels.

- Data rows (in <tbody>) must have a gray background.

- Do not summarize or sample the data. Show the raw rows exactly as they appear in the CSV.

##

In the LLM section, add a summarization prompt.



Summarization Prompt:

##

Based on the information through the chat history and/or tool call responses, construct a clear, professional, and user-friendly answer to address the user's original query.

construct an answer that addresses the user's original query.

Your answer should be directed towards the user and should be friendly, professional,

and concise. If you are not able to answer the user's question based on the tool calls or through the chat history,

you should let them know that you do not have access to that information.

--ANSWER REQUIREMENTS--

Your answer should be based only on the information available to you in the tool calling chat history. You should never invent or make up answers. If the question requires you to synthesize answers across several tool responses, you should do that. Note that some information may need to be summarized from across multiple places in a given object. Use your inference and semantic knowledge to derive the answer from the data without inventing anything.

--ADDITIONAL REQUIREMENTS--

* Do not mention tool calls, chat history, underlying operations, or system processes.

* Keep the response natural and conversational without unnecessary preambles.

* Provide a direct and engaging response without phrases like "Based on tool calls" or similar references.

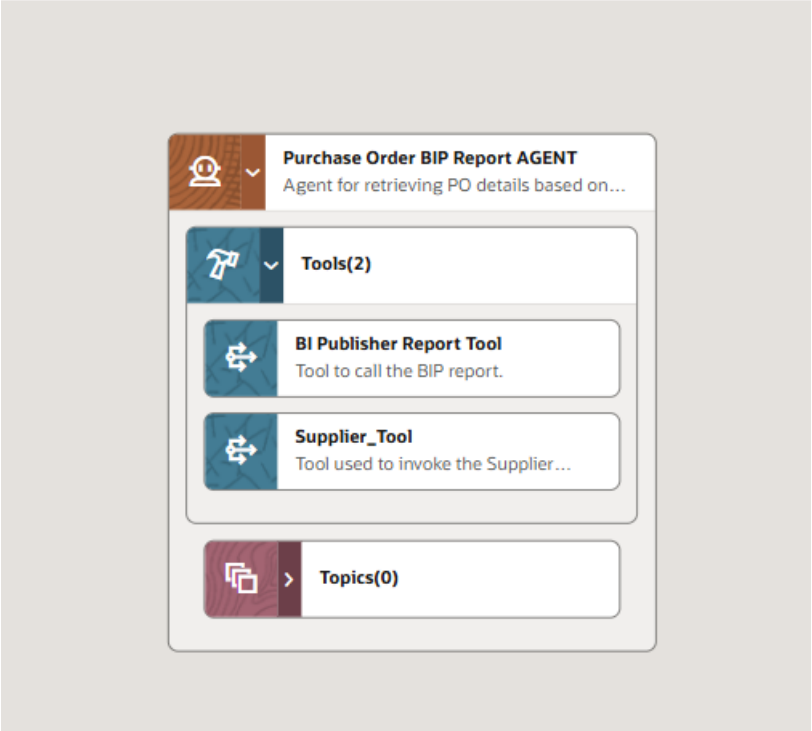
* Do not append any characters or trailing Periods (.) at the end of valid URLs.

* If link is generated by DeepLink tool do not remove from response.

Please show "PO data" in HTML table with headers in yellow and rows in gray.

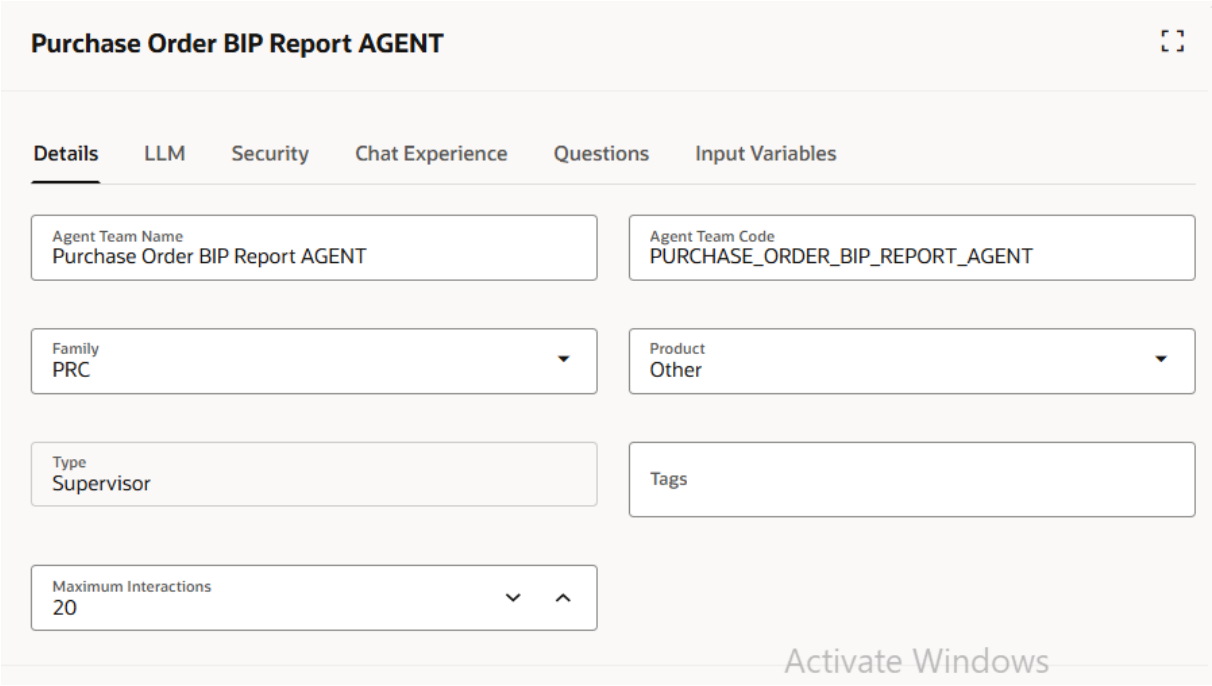
##

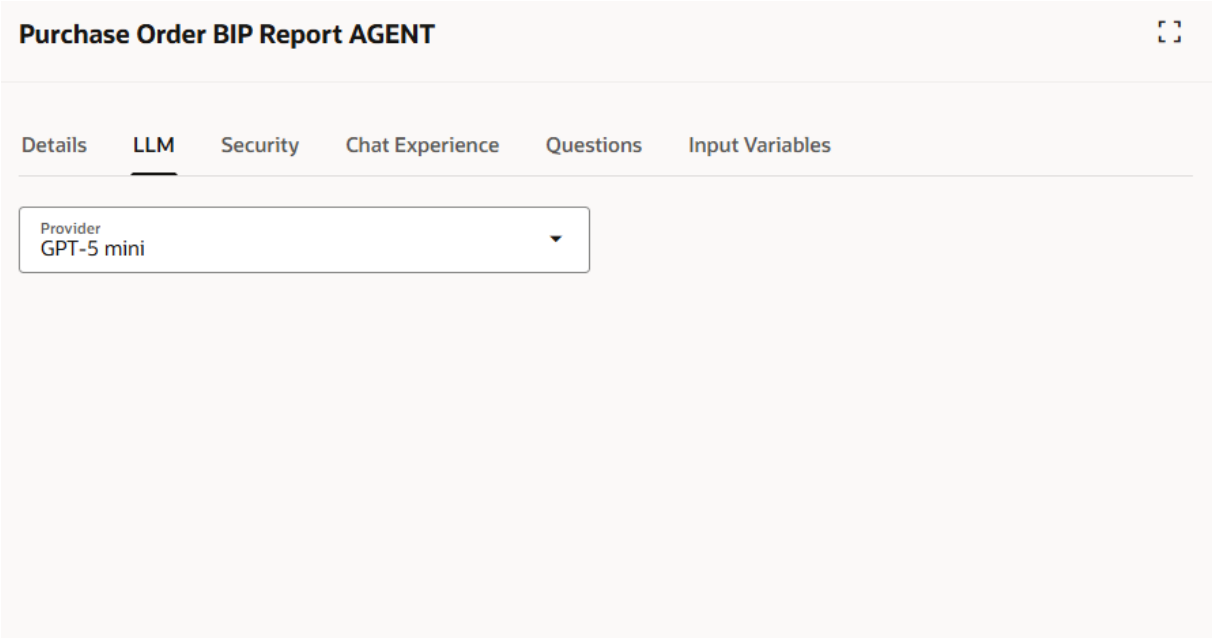
The Custom Agent will appear as follows after creation:



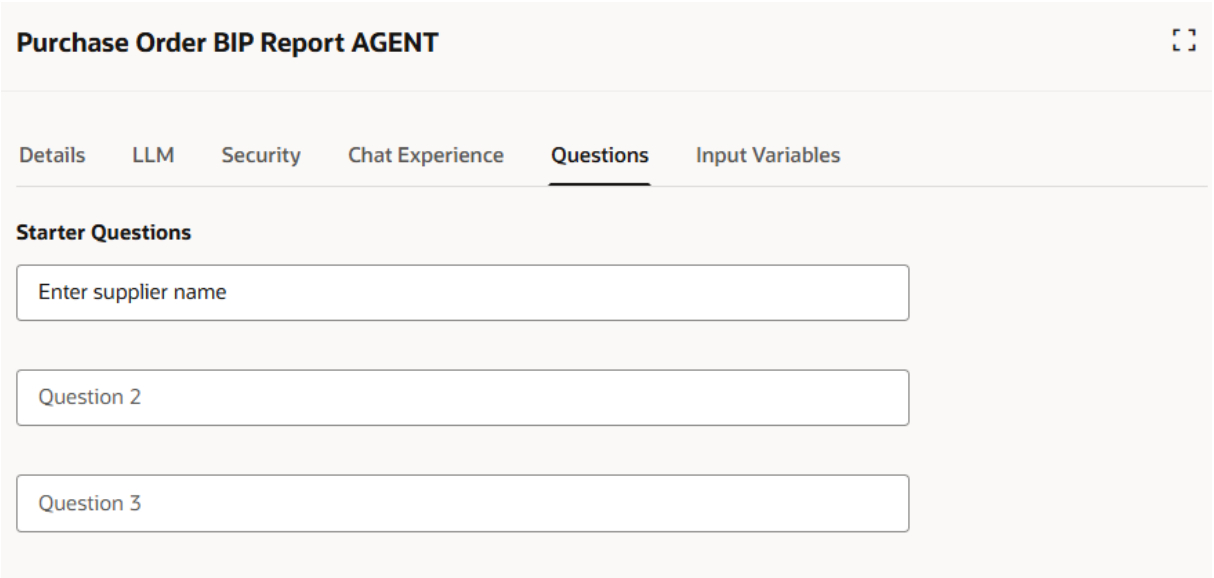
Agent Team:

Create an agent team with the type set to Supervisor.

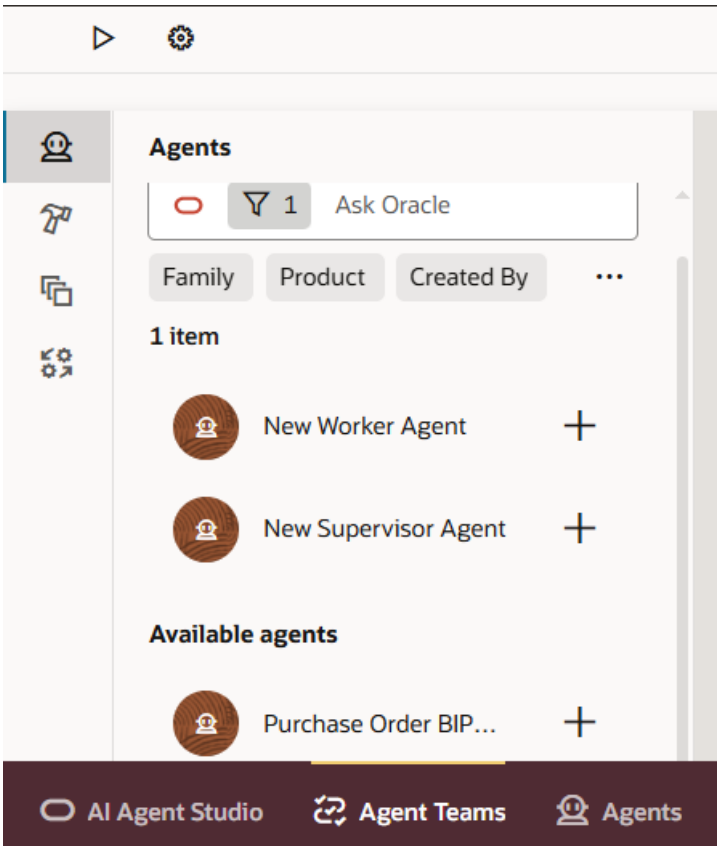




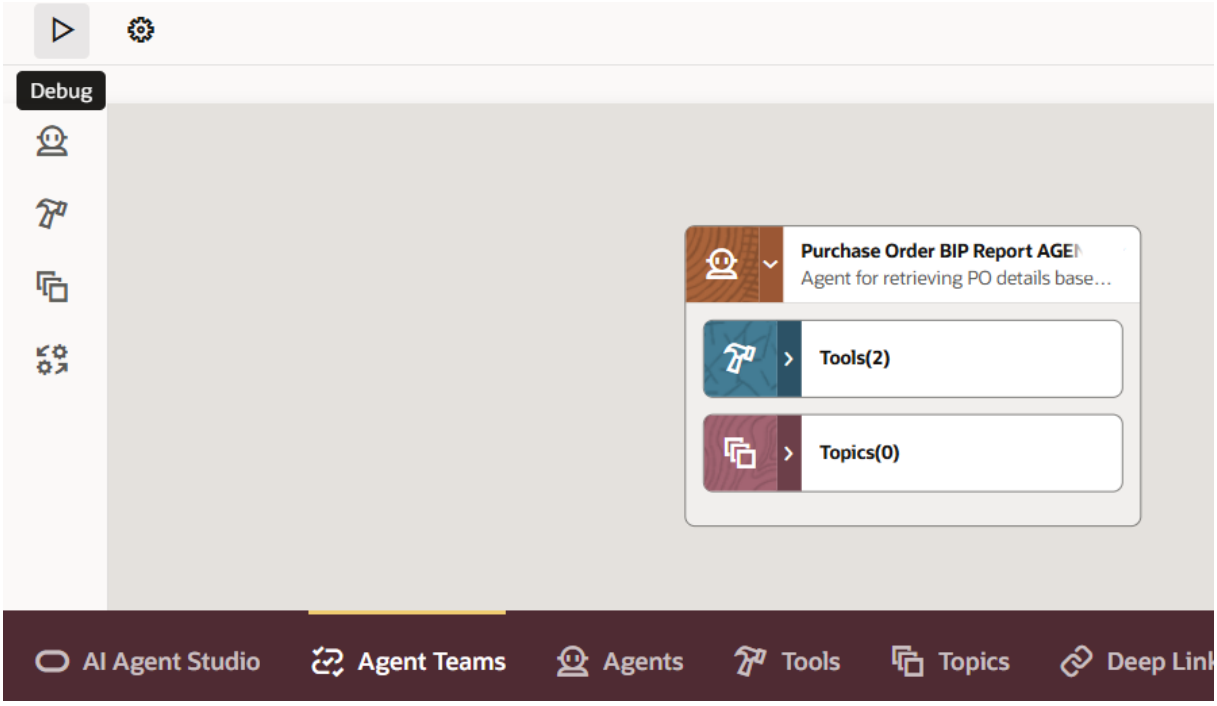
Add the start questions that you want the agent to show to the user.



Add the Custom Agent that you just created.



To test, click the Debug button as shown below.



Purchase Order BIP Report AGENT

How can I help you?

Enter supplier name

ATD

Activate Windows
Go to Settings to activate Windows.

Ask Oracle

Purchase Order BIP Report AGENT

LLM Call(1)

BI Publisher Report Tool

BI Publisher Report Tool

LLM Call(2)

LLM Call(3)

Here is the PO information for the supplier **ATD**:

PO_NUMBER	SUPPLIER	CURRENCY_CODE	ITEM_DESCRIPTION	PO_LINE_ID	REQ_BU_ID	PO_HEADER_ID	PRC_BU_ID	SOLD_TO_LE_ID	LINE_NUMBER	SUPPLIER_NAME
275	10314	EUR	Description accord	300000015402353	300000002286875	3000000015402352	300000002286875	300000001901094	1	ATD
			Descri	3000	3000	30000	3000	3000		

Activate Windows
Go to Settings to activate Windows.

Ask Oracle

Purchase Order BIP Report AGENT

NUMBER	PPLIER	RENCY_CODE	DESCRIPTION	PO_LINE_ID	REQ_BU_ID	PO_HEADER_ID	PRC_BU_ID	D_TO_LE_ID	_NUMBER	LIER_NAME
275	10314	EUR	Description accord	30000015402353	30000002286875	300000015402352	30000002286875	30000001901094	1	ATD
275	10314	EUR	Description accord	30000015402356	30000002286875	300000015402352	30000002286875	30000001901094	2	ATD
274	10314	EUR	Description accord	30000015402348	30000002286875	300000015402347	30000002286875	30000001901094	1	ATD
367	10314	EUR	Ligne 5 - Services	30000023295346	30000002286875	3000000232 95345	30000002286875	30000001901094	1	ATD
367	10314	EUR	Ligne 5 - Services	30000023295349	30000002286875	3000000232 95345	30000002286875	30000001901094	2	ATD
413	10314	EUR	Boite à outlets	300000252268	30000002286862	300000252260	30000002286875	30000001901094	1	ATD

Activate Windows
Go to Settings to activate Windows.

Ask Oracle